D06 · Network Design & 3PL Management

# 05 · Sustainable logistics: carbon footprint, green transport & circular supply chains

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

## 1 WHAT IT IS

Sustainable logistics measures and reduces the emissions and waste of transport, warehousing and packaging — using recognized methods such as the GLEC Framework and ISO 14083 — and redesigns flows for lower carbon and, often, lower cost.

## 2 WHY IT MATTERS

Transport emissions are a material part of most companies' Scope 3 footprint, and customers and investors increasingly ask for credible numbers. **Many of the strongest levers — consolidation, higher fill, modal shift to rail, fewer empty kilometers — also cut freight cost.** The precondition is measurement by lane and mode with a recognized method; claims without a baseline do not survive scrutiny.

## 3 HOW IT IS DONE

1. **Build the emissions baseline.** Per lane, mode and customer, using GLEC / ISO 14083 methodology and actual activity data.
2. **Rank levers by carbon and cost.** Start with the ones that reduce both: consolidation, fill rate, modal shift, network changes.
3. **Bring carriers and 3PLs in.** Emissions data and intensity targets in scorecards and contracts.
4. **Report consistently.** Same method every year, with intensity and absolute figures.

### IN PRACTICE

**Measure before you set targets**

A credible baseline is the prerequisite for any serious reduction program and for external claims.

**Prioritize levers that improve both emissions and cost**

Consolidation, fill-rate improvement, modal shift and network optimization usually sit in this quadrant.

**Include emissions expectations in 3PL and carrier governance**

What gets measured and discussed in QBRs gets managed. Sustainability that remains outside the commercial relationship remains aspirational.

### ILLUSTRATIVE CASE

*Illustrative case built from typical industry values — not data from a specific company.*

**Illustrative case — Consumer goods company setting a logistics emissions reduction target** — Baseline showed road freight dominating both cost and emissions.

| Lever | Emissions Impact | Cost Impact |
|---|---|---|
| Improved consolidation + higher fill rates | -9% | -6% freight |
| Modal shift of suitable volume to rail | -14% on shifted lanes | Neutral to slightly favorable |
| Carrier scorecard including emissions intensity | Gradual improvement | Slight premium on some lanes |
| Packaging / unitization improvements | -3% | Favorable |

**Result:** The company achieved a 17% reduction in logistics emissions intensity over three years while simultaneously reducing total freight cost. The initiatives that improved both dimensions were prioritized first.

### HOW IT IS MEASURED

**Logistics COe per Unit / per tkm**

Emissions intensity of transportation and distribution activities

*Benchmark: Track absolute and intensity; intensity is usually the more actionable management metric*

**% of Volume on Lower-Emission Modes**

Share of freight moved by rail, intermodal or other lower-intensity modes

*Benchmark: Depends on corridor suitability; the goal is to maximize appropriate shift*

### WHAT YOU WOULD USE

**Sustainable Logistics**

**EcoTransIT / NTMCalc / similar emission calculation engines** — Widely used for transport emission calculations aligned with recognized standards.

**Visibility platforms with emissions modules (e.g. project44)** — Increasingly offer per-shipment or per-lane emissions tracking integrated with operational data.

## 4 SO WHAT

**Measure with a standard, then cut what costs twice.** Baseline by lane with GLEC / ISO 14083, prioritize levers that save both carbon and money, and make carriers part of the target.

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